

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: OE&NA**

2. **Subject Name: Propulsion** **L-T-P: 3-0-0** **Credits: 3**

3. **Pre-requisites:**

4. **Syllabus and reference books:**

Syllabus:

- Introduction to Ship Propulsion
- Screw Propeller Geometry
- Propeller Theories: Momentum Theory, Blade Element Theory, Circulation Theory, Other Models for Propeller Action
- Laws of Similarity, Dimensional analysis, Open water characteristics, Methodical Series
- Hull-Propeller Interaction, Propulsive Coefficients, Efficiency
- Propulsion Model Tests, Ship Powering Performance Extrapolation
- Engine-Propeller Matching
- Propeller Cavitation
- Propeller Materials, Blade Strength
- Propeller Design Methods
- Controllable Pitch Propellers, Ducted Propellers etc.
- Unconventional Propulsors, Thrust Augmentation (Energy Saving) Devices.

Reference Books:

- (1) *'Marine Propellers and Propulsion'* by John Carlton, Butterworth-Heinemann Publisher.
- (2) *'Basic Ship Propulsion'* by J.P. Ghose and R.P. Gokarn, KW Publishers Private Limited.
- (3) *'Ship Resistance and Propulsion'* by A.F. Molland, S.R. Turnock, and D.A. Hudson, Cambridge University Press

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
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1.	Introduction to Ship Propulsion	2
2.	Screw Propeller Geometry	4
3.	Propeller Theories	4
4.	Laws of Similarity	6
5.	Hull-Propeller Interaction,	4
6.	Propulsion Model Tests	6
7.	Engine-Propeller Matching	3
8.	Propeller Cavitation	4
9.	Propeller Cavitation	3
Total number of hours		36

